◇ Grassmannian

Let V be a complex vector space of dimension n. The Grassmannian $G(k, V)$ is defined to be the set of k-dim linear subspaces of V.

* $G(k, n) := G(k, \mathbb{C}^n)$

Given a k-plane Λ in $\mathbb{C}^n$, we may represent Λ by a set of k row vectors in $\mathbb{C}^n$ spanning Λ, i.e., by a $k \times n$ matrix

$$\begin{pmatrix} v_{11} & - & - & \cdot & v_{1n} \\ v_{21} & & & & v_{2n} \\ \vdots & & & & \vdots \\ v_{k1} & & & & v_{kn} \end{pmatrix}$$

of rank k. Clearly any such matrix represents an element of $G(k,n)$ and any two such matrices A, A' represent the same element of $G(k,n)$ iff $A = gA'$ for some $g \in GL_k$

For every multiindex $I = \{i_1, \dots, i_k\} \subset \{1, \dots, n\}$ of cardinality k, let $V_{I^\circ} \subset \mathbb{C}^n$ be the $(n-k)$-plane in $\mathbb{C}^n$ spanned by the vectors $\{e_j : j \in I\}$, and let $U_I = \{\Lambda \in G(k,n) : \Lambda \cap V_{I^\circ} = \{0\}\}$

U_I is just the set of $\Lambda \in G(k,n)$ s.t. the I-th $k \times k$ minor of one, and hence for any, matrix repre. for Λ is nonsingular. Any $\Lambda \in U_I$ has a unique matrix repre. Λ^I whose I-th $k \times k$ minor is the identity matrix, e.g. any $\Lambda \in U_{\{1,\dots,k\}}$ can be represented uniquely by a matrix of the form

$$\begin{pmatrix} 1 & & 0 & \vdots & \cdots & \cdot \\ & \ddots & & \vdots & & \vdots \\ & & \ddots & \vdots & & \vdots \\ 0 & & & 1 & \cdots & \cdot \end{pmatrix}$$

Conversely, any $k \times n$ matrix of the form above represents a k-plane $\Lambda \in U_I$; thus the $k(n-k)$ entries of the I°th $k \times (n-k)$ minor $\Lambda^I_{I^\circ}$ of Λ^I give a bijection of sets

$$\varphi_I : U_I \longrightarrow \mathbb{C}^{k(n-k)}$$

for each I.

△ 1D solvable model

$$H(t) = \frac{\hbar\omega_0}{2} \sin(\omega t)\, \sigma_x + \hbar\omega_0 \sin^2\left(\frac{\omega t}{2}\right) \sigma_y$$

$T = \frac{2\pi}{\omega}$

$$= \hbar\omega_0 \begin{pmatrix} 0 & \frac{\sin(\omega t)}{2} - i\sin^2\left(\frac{\omega t}{2}\right) \\ \frac{\sin(\omega t)}{2} + i\sin^2\left(\frac{\omega t}{2}\right) & 0 \end{pmatrix}$$

instantaneous energy eigenvalues $E_X = X \hbar\omega_0 \sin\frac{\omega t}{2}$

The two bands cross at $t=0$, $t=T$

$$G(\omega t) = \begin{pmatrix} 0 & e^{-i\omega t} \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \cos(\omega t) - i\sin(\omega t) \\ 1 & 0 \end{pmatrix}$$

$$G^{-1}(\omega t) = \begin{pmatrix} 0 & 1 \\ e^{i\omega t} & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \cos(\omega t) + i\sin(\omega t) & 0 \end{pmatrix}$$

$$G H G^{-1} = \begin{pmatrix} \frac{\sin(\omega t)}{2} - i\sin^2\left(\frac{\omega t}{2}\right) & 0 \\ 0 & \frac{1}{2}\sin(\omega t)\cos(\omega t) + i\sin^2\left(\frac{\omega t}{2}\right)\cos(\omega t) - \frac{i}{2}\sin^2(\omega t) + \sin^2\left(\frac{\omega t}{2}\right)\sin(\omega t) \end{pmatrix} G^{-1}$$

$$= \begin{pmatrix} 0 & \frac{\sin(\omega t)}{2} - i\sin^2\left(\frac{\omega t}{2}\right) \\ \begin{array}{l} \frac{1}{2}\sin(\omega t)\cos^2(\omega t) + i\sin^2\left(\frac{\omega t}{2}\right)\cos^2(\omega t) \\ \cancel{-\frac{i}{2}\sin^2(\omega t)\cos(\omega t)} + \cancel{\sin^2\left(\frac{\omega t}{2}\right)\sin(\omega t)\cos(\omega t)} \\ \cancel{+\frac{i}{2}\sin^2(\omega t)\cos(\omega t)} - \cancel{\sin^2\left(\frac{\omega t}{2}\right)\sin(\omega t)\cos(\omega t)} \\ +\frac{1}{2}\sin^3(\omega t) + i\sin^2\left(\frac{\omega t}{2}\right)\sin^2(\omega t) \end{array} & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{\sin(\omega t)}{2} - i\sin^2\left(\frac{\omega t}{2}\right) \\ \frac{\sin(\omega t)}{2} + i\sin^2\left(\frac{\omega t}{2}\right) & 0 \end{pmatrix}$$

$$= H$$

$$G^2 = \begin{pmatrix} 0 & e^{-i\omega t} \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & e^{-i\omega t} \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} e^{-i\omega t} & 0 \\ 0 & e^{-i\omega t} \end{pmatrix}$$

△ Classification of relative homotopy group

d-dim : $BZ \overset{\text{homeomorphic}}{\cong} T^d$

Assume the band crossing occurs at a single point in the momentum space $\{w\}$

We assume there are n occupied, m unoccupied Bloch bands for each momentum.

Consider $T^d \setminus \{w\}$, define Q-matrix

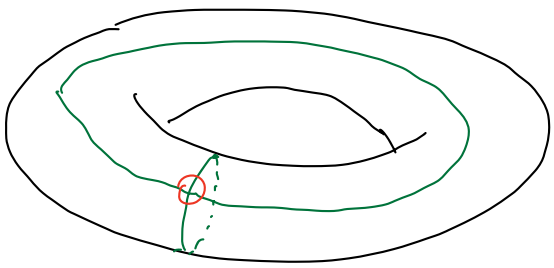
$$Q(\vec{k}) = 2P(\vec{k}) - \mathbb{1}, \quad \text{where}$$

$$P(\vec{k}) := \sum_{a=1}^{n} |u_a(\vec{k})\rangle\langle u_a(\vec{k})|$$

↳ occupied Bloch wave-function

对于一个特定系统，$Q(\vec{k})$ defines a continuous map

$$T^d \setminus \{w\} \longrightarrow M \text{ (some manifold)}$$



$\cup$ submanifold

X

$$\circ \cong S^{d-1}$$

$$G_{\mathbb{C}}(m, n+m) \cong \frac{U(m+n)}{U(m) \times U(n)}$$

$$\Longrightarrow \pi_d(M, X)$$

在 degenerate point 附近，由于 parity symmetry 仍存在，Q 具有 off-diagonal 结构，因此 $\{0\} \longrightarrow U(n)$